Bruin Home Solutions
May 11, 2020

Project Proposal

Sunset Aquaponics Garden

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Executive Summary

The purpose of this proposal is to outline our design for an aquaponics system to be integrated at the UCLA Sunset Canyon Recreation Center. This proposal will address several sustainability goals that UCLA has put in place, including sustainably sourcing 20% of UCLA's food and achieving zero waste by 2020. Since 42% of UC students experience food insecurity, our system, which will generate an estimated 2000 pounds of produce in each year of operation, will greatly help to feed food insecure students, especially those at UCLA by supplying the food closet in the Student Activities Center. By design, crop cultivation by aquaponics has tremendous advantages over both conventional farming methods and hydroponics systems because it eliminates nutrient waste leakage to the surrounding environment.

In terms of design, the Sunset Aquaponics Garden will consist of four 800 gallon cylindrical stock fish tanks, each stocked with a mixture of koi and goldfish. The water will flow out of these tanks into the mechanical and biological filters, and then it will enter the plant grow beds. The four fish tanks will flow into ten plant grow beds, which will each have floating polystyrene rafts on which plants will grow. Initially, we will grow microgreens and larger leafy greens. After achieving success, we will transition to grow larger varieties of fruits and vegetables including chard, squash, and strawberries. From the plant beds, the water will flow to a sump, where it will collect and then be pumped back to the fish tanks.

The Sunset Aquaponics Garden is innovative, due in part to the design of the mechanical filter system which almost entirely automates its own cleaning and maintenance process, as outlined below in the section entitled "System Design." Thus, our system presents a unique opportunity to measure quantitatively the effects of these filtration systems on aquaponics produce output. These quantitative results are generated by an extensive network of sensors, which will provide data output continuously to an interactive data display web application that allows for real-time adjustments to be made to the system to optimize growth output in terms of dry weight of harvested produce. Our system will also generate all of its own electricity with a 5 kW solar panel array, which will power the system during the day and store energy in a battery network to be used at night.

The Sunset Aquaponics Garden presents a rare and unique opportunity to engage the UCLA community in cutting-edge sustainability research and an initiative to generate produce in support of the food-insecure population at our university. We hope that our system will serve as a seed system for other aquaponics systems at UCLA and other UC campuses to help combat similar sustainability issues, as the Sunset Aquaponics Garden will be just one of an estimated 1,000 school aquaponics systems in the United States, further highlighting the need for the implementation and novel design of the Sunset Aquaponics Garden.

1. Introduction:

Bruin Home Solutions (BHS) is a student-run, non-profit organization. Our mission is to support UCLA in achieving ambitious sustainability goals such as sustainably sourcing at least 20% of the food on campus and eliminating all waste by 2020.

The aquaponics farm is an initiative to research food growth using less resources and producing less waste than traditional farming. By design, it integrates aquaculture with hydroponics. The system discussed in this proposal is closed-looped, meaning all wastes generated by the fish are uptaken by the plants as nutrients, such that no waste leaves the system and enters the surrounding environment, mimicking a natural ecological system that is symbiotic for both fish and plants. Additionally, since the system will grow food on campus, it will eliminate pollution from transporting food from farm to table. Our team will focus on innovative system design to generate food for UCLA students in the on-campus dining halls, as well as add fresh produce to the food closet at the student activities center to feed UCLA's food-insecure students. Furthermore, data collection will be integrated with system automation, and this data will be wirelessly uploaded to an interactive data display web application. The purpose for collecting data is two-fold: it enables us to share our research with the public and view and analyze our data in real time. Thus, we will be able to adjust and optimize the efficiency of the system as required.

The total estimated budget for the system is $21,046, and we plan to have the system fully operational by March 15, 2021. This projected cost and timeframe, along with the system design, was developed by our team with the assistance of Brian Hsiao, our graduate student mentor, and Zack Osborn, a student research engagement specialist with experience in aquaponic and wastewater design, part of the Bioregional Center at UCSD. Their experience and expertise with aquaponics, combined with their enthusiasm towards our project, demonstrates that the Sunset Aquaponics Garden is feasible and will achieve success. This proposal discusses the design, implementation, maintenance, and budget of the Sunset Aquaponics Garden, which would ideally be erected at the Sunset Canyon Recreation Center. Overall, this system will help to bridge the gap between the UCLA population and its food source.

2. Needs/Problems:

A majority of the food we consume is transported an average of 889 miles from farm to table in the United States. This method of farming and transportation pollutes our air, soil, and water. This conventional model of food production leads to food waste in distribution, nutrition loss, and many negative environmental impacts such as decreased disease resistance in monocrops.

A UC systemwide survey revealed that 42% of students experience food insecurity. Currently, most of the food available for food-insecure students is canned and non-perishable dry goods, which means the diets of food-insecure students are often lacking in fresh fruits and vegetables. The efficient yield of 2,000 lbs per year of organic, nutrient-rich vegetables by the

Sunset Aquaponics Garden would be a healthy source of food for hungry students. The food could be served directly in the dining halls here, potentially lowering the cost of the meal plan. More importantly, the food could be donated to the food closet and made directly available to food-insecure students. Additionally, we can use a portion of the produce generated by the Sunset Aquaponics Garden to supply the food closet located at the student activities center.

The massive and ever-increasing population in cities such as Los Angeles demands a progressively larger food supply. Currently, most of our food originated outside of the city and needs to be transported here, causing pollution. This research project has substantial practical application for large scale, urban food cultivation that will become increasingly economically advantageous as the population continues to grow, and aquaponics technology begins to fall in price.

3. Partnership Concepts:

In order to achieve our education goals, as listed below in the “Goals” section, we plan to create a student organization dedicated to the maintenance and research of the Sunset Aquaponics Garden. This organization will be led by expert advisors and will be open to all UCLA undergraduate and graduate students. Members will assist in system maintenance and in developing system updates, and will also serve as key ambassadors in the education of future environmentalists.

Furthermore, because our proposal will integrate the system into the Sunset Canyon Recreation Center at UCLA, we aim to incorporate our system into the Sunset Canyon Recreation Center redesign for the 2028 Olympics, which are being held in Los Angeles. Success in this endeavor will allow the Sunset Aquaponics Garden to champion sustainability at the event and to demonstrate that UCLA and the greater Los Angeles community are global leaders in sustainability.

4. Goals:

The Sunset Aquaponics Garden is being built with the following five core ideas in mind: sustainability, food security, pollution reduction, research, and education. Below, we describe the benefits of each goal and how we plan on measuring these objectives.

Sustainability: Building a sustainable system means building an efficient operational workflow that conserves resources and reduces costs. In terms of sustainability, we are considering water, energy and material usage.

Food Security: Food security issues affect a large portion of UC students; this system will help to alleviate that problem by providing a consistent source of fresh fruits and vegetables to the UCLA food bank.

Pollution Reduction: All the food that comes through the UCLA dining halls is imported from central farming operations. This produces a huge amount of pollution through the

production of the food and transporting food to the school. If we can effectively begin to replace that food with food homegrown here, we can greatly reduce that pollution.

Research: Aquaponics is still a relatively novel concept. We will be somewhat pioneering this design and we hope to have others follow in our footsteps. The primary method to accomplish this will be sharing our design, all operational and unforeseeable problems, and collected data. We will measure the levels of nitrogen compounds in the system, pH levels, electrical conductivity (EC), dissolved oxygen (DO), produce yield, energy usage, and the effectiveness of our filtration systems (indicated by total sediment build up in grow beds). Our research will focus primarily on increasing the automation of the system in order to decrease the necessary labor input to decrease cost and create a greater commercial viability.

Education: As a world renowned academic institution, we should not limit our experiential learning to the classroom. In courses such as CHS 131, this aquaponics system has the potential to teach students the intersection of cutting edge technology and sustainable food systems. Additionally, this system can be used as a learning tool for UCLA Extension certificate programs, as well as with academic programming for youth groups who use sunset rec for their summer programming. With thousands of kids visiting our facility over the summer, we could use this system as an opportunity for teaching future environmentalists. Cultivating future environmentalists is a key to achieving a sustainable future and it would be a shame to miss this opportunity to educate our community.

5. Maintenance:

Following the construction of the Sunset Aquaponics Garden, partial and whole-system maintenance will be required. Regular maintenance will be automated, the mechanical filter system backflows itself automatically, fish are fed automatically, and all water nutrient data is collected and analysed automatically by the computer control system and any nutrient deficiency is automatically compensated for by injection of aqueous nutrient salt solution. Monthly maintenance will consist of restocking the automatic fish feeding system and checking the mechanical filter backflow system. Sensors will also be checked to make sure the dispensers are stocked.

Full system maintenance will be carried out every three months. This entails restocking plants. We will also remove the biofilter and other system parts for cleaning. Hydrogen peroxide will be used to clean and sterilize the system components. This maintenance will be performed by a group of students from Bruin Home Solutions who will be using the system to conduct research.

6. System Design:

The system (Figure 1) comprises four 800 gallon stock tanks holding koi and goldfish. The tanks are outfitted with automatic fish feeders which drop fish feed into the tanks at a sustained daily rate so human interaction with the system is minimized on a daily basis. There is an outlet pipe on each of the tanks 2 inches below the water line. Water flows into these pipes

which conjoin into a single pipe outfitted with a Libelium automated water quality sensor configured for nitrogen compounds, pH, temperature, dissolved oxygen, and electrical conductivity. This data is recorded and sent to an arduino microcontroller. If pH or micronutrient levels are too high or too low, the arduino triggers automated droppers of sodium hydroxide and hydrochloric acid to drip into the tanks, correcting the pH to proper level.

From the sensor section water flows into the first stage mechanical filter (Figure 2). This first filter comprises a T-shaped section of PVC with automated ball valves at the end of each arm of the T and the bottom of the vertical section. In the downstream arm of the T there are disks of increasingly fine plastic mesh which trap suspended solids. After the mesh and before the ball valve there is a connector for a garden hose. During normal operations the ball valve at the bottom of the vertical section is closed and the ball valves in each arm of the T are open allowing water from the tank to flow through the mesh on its way to the biofilter. Several times a week the arduino board which controls the entire system closes the inline flow valves at each end of the T, stopping flow to and from the filter and isolating it from the rest of the system. After isolating the filter, the valve at the bottom of the T opens and allows the water and trapped solids in the filter to drain out, and then the arduino opens the valve to the garden hose located behind the mesh layers. This high pressure water backflows through the mesh, removing any trapped solids and cleaning the mesh layers. It also prevents biosolid buildup, which could clog the filter. The hose valve is closed, then the valve at the bottom of the T is closed, and then the inline flow valves are opened and normal operation resumes. This design ensures almost no human interaction is necessary in cleaning the filter, reducing labor requirements. The water is cleaned of particulate matter and then flows into the biofilter.

The biofilter (Figure 3) is a 55-gallon drum filled with layers of sponge-like, plastic mesh of varying size. This mesh supports nitrifying bacteria which convert the ammonia from the fish waste into nitrates, which the plants can use as fertilizer. From the biofilter the water flows into 10 PVC pipes which feed into the grow beds which are 4’x8’x12” formed out of 2”x12” lumber with a ¾” plywood bottom lined with pond liner. In each bed two 4’x4’ polystyrene sheets float on top of the water, and holes cut in each sheet hold the plants and allow their roots to hang into the nutrient rich water in the grow bed.

Plants will be germinated in rockwool cubes and then transplanted into the system. At the end of each grow bed there is a PVC pipe at the waterline through which water flows out and into the final sump tank, where four 240 GPH (gallons per hour) aquarium pumps continuously pump water from the sump tank back into the fish tanks. In each grow bed there is an airstone which keeps the dissolved oxygen concentration high enough to keep the plants healthy. Four air stones are placed in each fish tank to keep the fish healthy and discourage the buildup of anaerobic bacteria, which can produce methane and harm the fish. All air stones are connected to a bank of four 850-GPH 18W air pumps. The system is powered by a 5 kW solar array, which produces more power than the system needs during the day and thus allows for energy storage in a 3 kW-h battery, which will power the system at night. Using solar power to

power an aquaponics system is a relatively new concept and is one of the main research points in our proposal.

The entire system is designed to be modular and more fish tanks and mygrow beds can easily be integrated to expand the system. In addition, in the future, specially designed grow beds can be added to the system to grow specialty crops, like tomatoes or strawberries, both of which grow exceptionally well in aquaponic systems but are not well suited for floating raft farming. Floating raft farming is more useful for leafy green crops such as lettuce, spinach, bok choy, and arugula, and thus we will grow those initially.

7. Research:

As discussed above, aquaponics is a relatively new and generally uncommon practice. As a result, one of our main goals in designing and building this system will be research. The first component of this research goal is monitoring the chemistry of the system, including pH, temperature, electrical conductivity (EC), dissolved oxygen (DO), and nitrogen compounds, including ammonia, nitrite, and nitrate. Measuring these values (and uploading them to an interactive data display web application) will allow us to adjust them in real time so that they can be optimized to maximize system efficiency. Furthermore, we will measure plant output in terms of both dry weight of crop yield and grow cycle length. We anticipate that the length of grow cycles will be shorter than that of traditional farming, while the system produces a higher overall dry yield.

Additionally, because both our mechanical and biological filters are novel designs, so preliminary prototyping and testing will be done in order to ensure that both filters operate as efficiently as possible and require as little maintenance as possible. Also, our daytime energy will be produced entirely by solar panels, which is very atypical of aquaponics systems. Thus, this adds an additional research element to our system, as our system will becomes a case study in the effectiveness of using photovoltaic solar energy to run a medium-scale aquaponics system.

8. Key Personnel:

Client	**UCLA Housing**
Sponsor	The Green Initiative Fund (TGIF)
Project Managers	Shreesh Karjagi, Conor McCaulley
Team	Jay Uppuluri, Patrick Wilson
Mentors	Brian Hsiao, Zack Osborn
BHS President	Riley Kuffner

9. Materials/Budget:

(Figure 4)

Material	Dimensions	Units	Cost ($)
Fish tank	800 gallon	4	3200
Lumber	2" x 12" x 8'	36	360
Pond liner	7' x 10'	12	420
Plywood	4’ x 8’x 3/4”	12	650
Charlotte pipe	3" x 10'	16	160
PVC ball valve		12	1000
Motorized b. valve	1/2"	4	120
Arduino board		1	20
Automatic fish feeder		4	96
Libelium water sensor		1	2000
Koi		20	3800
Goldfish		60	150
PVC couplers	3"	30	60
PVC pipe elbow	3"	30	90
Matala green filter pad	39.5" x 24" x1.5"	10	400
240 GPH pump		4	100
Airstone disk		40	300
Air pump		4	120
5 kw solar panels		20	4000
3 kwh Battery		1	3000
Lettuce seeds	5 pounds	1	110
Kale seeds	2 pounds	1	60
Spinach seeds	3 pounds	1	270
Total			**21046**

10. Timetable:

	Description of Work	Start and End Dates (Tentative)
Phase One: Prototyping	Build and test mechanical and biological filtration systems, automation systems, and sensors	**Start:** (10/01/2019) **End:** (11/01/2019)
Phase Two: Assess Site	Assess what needs to be done to prepare site for system installation	**Start:** (11/01/2019) **End:** (11/04/2019)
Phase Three: Break Ground	Level ground at the site to prepare for system installation	**Start:** (11/04/2019) **End:** (11/12/2019)
Phase Four: Begin Installation	Install fish tanks and grow beds at the site	**Start:** (11/12/2019) **End:** (12/5/2021)
Phase Five: Install Filtration Systems	Install mechanical and biological filters, as well as pumps and sump. Complete installation.	**Start:** (1/10/2021) **End:** (1/24/2021)
Phase Six: Fill System	Add water and fish to the system. Germinate plant seeds.	**Start:** (1/24/2021) **End:** (1/27/2021)
Phase Seven: Plant Produce	Transfer germinated plant seeds into the system. Operation of system begins.	**Start/End:** (2/02/2021)
Phase Eight: First Harvest	Harvest first crop of produce.	**Start/End:** (3/15/2021)

11. Appendix:

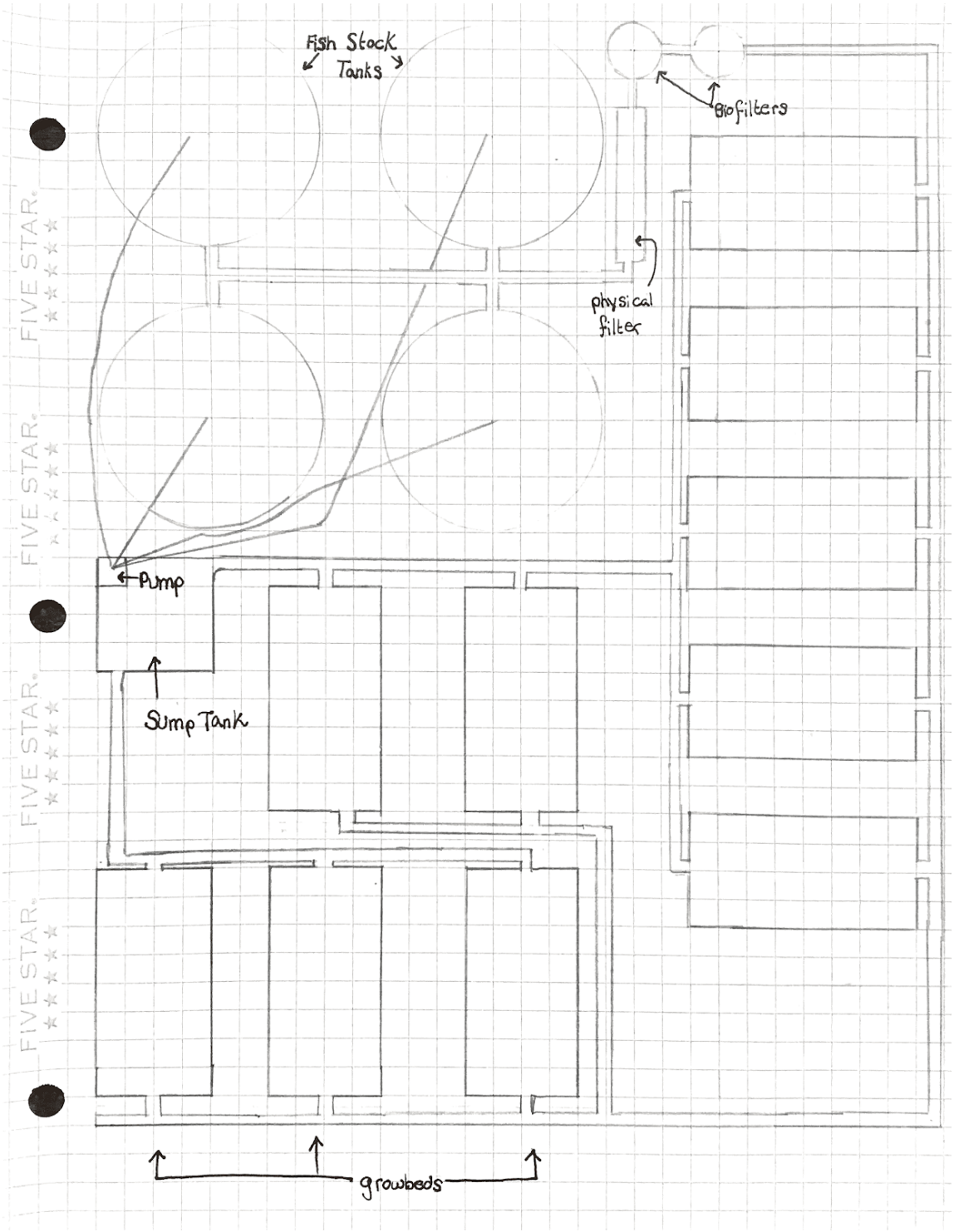

Figure 1. Overall System Layout

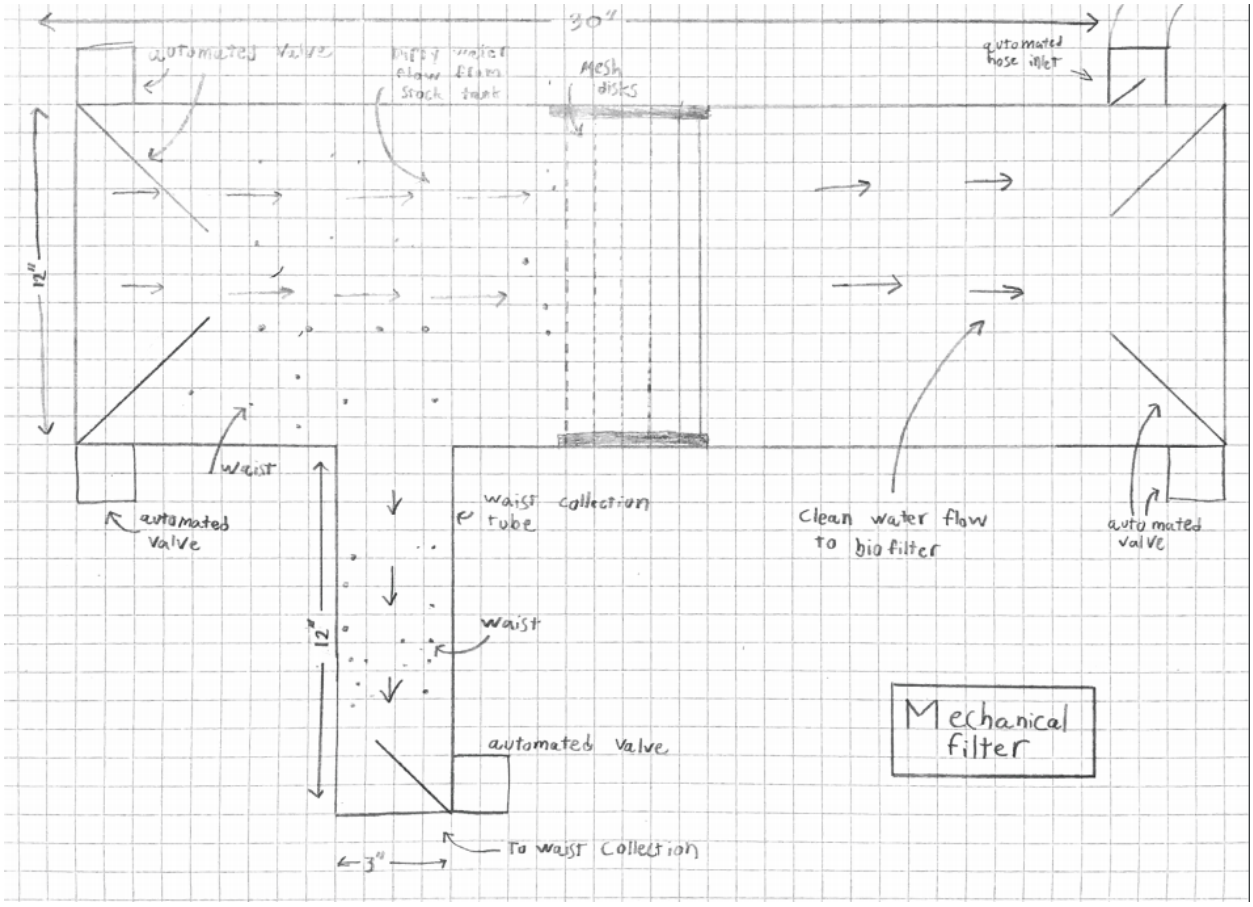

Figure 2. Mechanical Filter

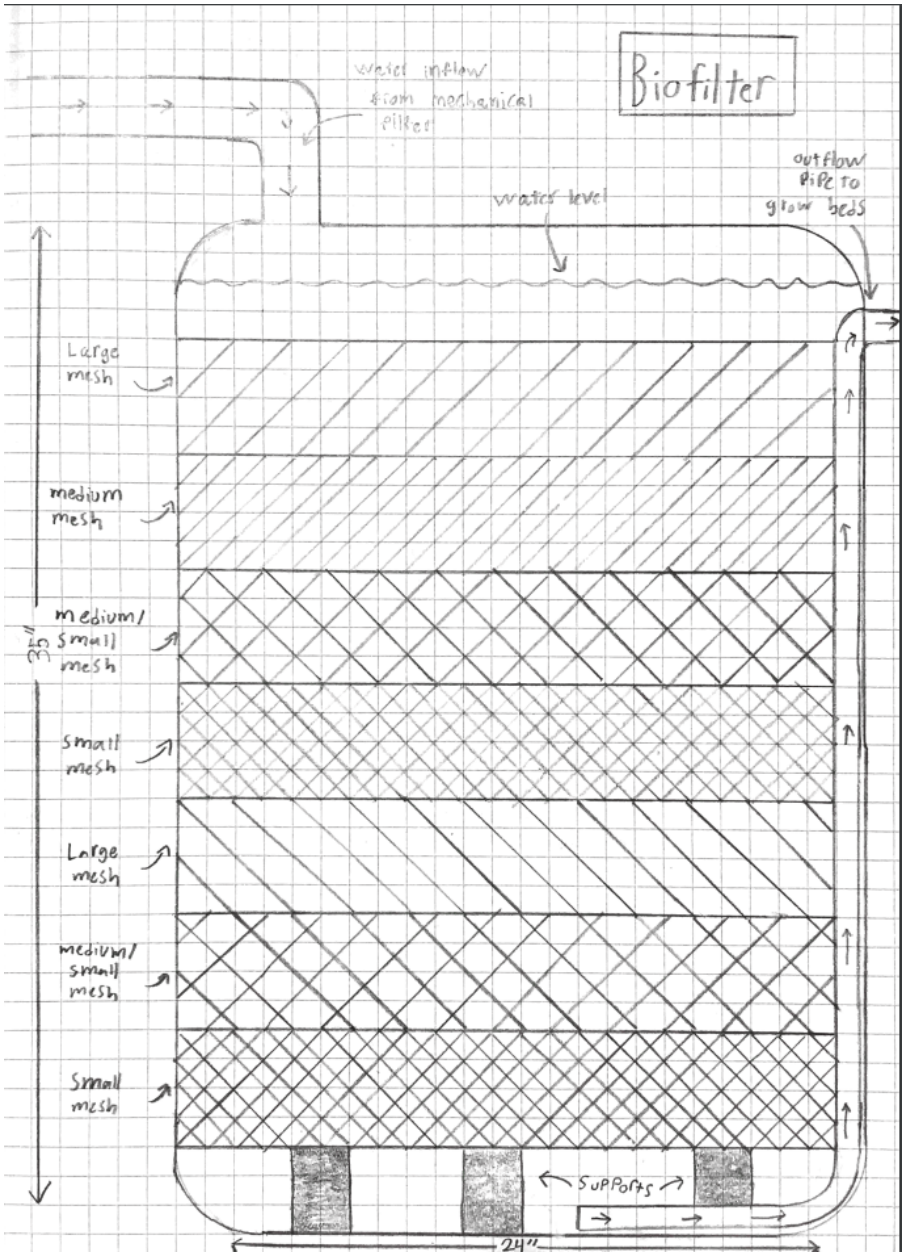

Figure 3. Biofilter

Material Sources (hyperlinked)	
Fish tank	PVC pipe elbow
Lumber	Matala green filter pad
Pond liner	240 GPH pump
Plywood	Airstone disk
Charlotte pipe	Air pump
PVC ball valve	5 kw solar panels
Motorized b. valve	3 kwh Battery
Arduino board	Lettuce seeds
Automatic fish feeder	Kale seeds
Libelium water sensor	Spinach seeds
Koi	Arugula seeds
Goldfish	Koi food
PVC couplers	

Figure 4. Materials